

Iron Deficiency Anaemia in Adults: A Primary Care Perspective

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Abstract

Iron deficiency anaemia (IDA) is the most common cause of anaemia worldwide and a frequent presentation in primary care settings. Identification typically relies on a complete blood count showing a low haemoglobin level, supported by iron studies. Common contributing factors include dietary insufficiency, chronic blood loss, and impaired absorption. This narrative review summarises widely accepted approaches to recognition, investigation, and conservative management of IDA in adults, with particular attention to settings where laboratory access may be limited [1, 2]. The review draws on publicly available patient-education and primary-care reference material and is intended as a clinician-facing summary rather than a clinical guideline.

Introduction

Anaemia affects a substantial fraction of the global population, with iron deficiency being the predominant cause across most age groups [3]. In low- and middle-income settings, the burden is particularly pronounced among women of reproductive age, infants, and young children [4]. Primary care clinicians are typically the first point of contact for patients presenting with non-specific symptoms such as fatigue, breathlessness on exertion, pallor, palpitations, hair thinning, or reduced exercise tolerance [5].

Early identification and appropriate investigation are essential. Iron deficiency is not itself a final diagnosis but a finding that may signal an underlying process - most commonly inadequate dietary intake, menstrual or other blood loss, or impaired gastrointestinal absorption [6]. In older adults the threshold for considering an underlying gastrointestinal source of bleeding is lower, and appropriate referral is advised [7].

This paper does not replace clinical assessment. All management decisions, including whether to investigate further and whether to initiate supplementation, should be made by a qualified clinician with access to the patient's full history.

Methods

This is a narrative review of publicly available primary-care reference material and patient-education resources from established health-information sources [1, 2, 4]. No new patient data were collected and no individual patient consent was required. Sources were selected for their general accessibility and their focus on the primary-care setting rather than tertiary or specialist contexts. Where sources disagreed, the more conservative recommendation appropriate for non-specialist settings was preferred.

Results

Across the reviewed sources, common presenting features reported in adults with iron

deficiency anaemia included generalised fatigue, reduced exercise tolerance, dizziness, occasional headaches, palpitations during exertion, and (in more advanced cases) brittle nails or hair thinning. Pallor of the conjunctivae or palmar creases may be observed on examination but is not a sensitive sign [5].

Laboratory findings most consistently reported were a haemoglobin level below the reference range printed on the laboratory report (which varies by age, sex, and altitude), reduced mean corpuscular volume in established cases, and a low ferritin level reflecting depleted iron stores [6, 8]. Reviewed sources stressed that a single out-of-range value should be interpreted in the context of the patient's clinical picture and prior trend, and that a repeat test is often appropriate before further investigation [9].

Investigation of the underlying cause should consider dietary intake, menstrual losses in women of reproductive age, possible gastrointestinal blood loss in older adults, and conditions affecting absorption such as coeliac disease in appropriate clinical contexts [7, 10].

Discussion

Management generally combines investigation of the underlying cause with iron repletion supervised by a clinician. Reviewed patient-education sources consistently advise discussing any planned supplementation with a clinician before starting, because inappropriate iron loading can be harmful in certain conditions and because the right form, route, and duration of supplementation depend on the individual [2, 8].

Dietary advice is appropriate as an adjunct. Iron-rich foods commonly mentioned in patient-education material include lean meats, pulses (lentils, beans), dark leafy green vegetables, fortified cereals, and dried fruit [1, 4]. Co-consumption of vitamin-C-rich foods with iron-containing meals is widely described as supporting absorption, while tea and coffee taken with meals are widely described as reducing absorption [10]. These are general dietary considerations and not a substitute for clinical management of significant deficiency.

Patients should be counselled to seek prompt review if they develop concerning symptoms · such as black or tarry stools, frank rectal bleeding, severe or worsening shortness of breath, chest discomfort, fainting, or symptoms that rapidly worsen · and to take their laboratory reports with them to any appointment so the clinician can review actual values and reference ranges [5, 9].

Conclusion

Iron deficiency anaemia remains a common and important primary care diagnosis. A structured approach to recognition, investigation of the underlying cause, and clinician-supervised management · combined with appropriate follow-up · is essential. Patient education should emphasise the value of bringing laboratory reports to consultations, reporting concerning new symptoms promptly, and discussing supplementation with a clinician rather than self-prescribing.

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